Prompt 4 - trial 3 (seed 1240)

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train two different GNN models on a compound activity prediction dataset and compare their performances. Select a compound from the test set correctly predicted as active by both models. Explain the predictions using EdgeSHAPer and highlight the differences between the learning characteristics of the models showing relevant visualizations.

Assistant

I'll help you train two different GNN models, compare their performances, and perform EdgeSHAPer explanations on a consensus compound. Let me start by exploring the available datasets. 🔍

(Calling tool `find_datasets`)

Assistant

Great! I found some ChEMBL activity datasets. Let me load a binary activity dataset and proceed with training two GNN models. I'll use the `chembl_activity_data_P14416_binary.csv` dataset. 💉

(Calling tool `load_dataset`)

Assistant

Let me check the correct label column name:

(Calling tool `load_dataset`)

Assistant

Excellent! The dataset has 8,348 compounds. Now let me split it into train/validation/test sets:

(Calling tool `split_dataset`)

Assistant

Dataset split complete! 🎯 Now let me prepare the GNN datasets for training:

(Calling tool `prepare_gnn_dataset`)

Assistant

Excellent! GNN datasets prepared. Now let me train two different GNN models - **GCN** and **GAT** - which have different architectures and learning characteristics:

🧬 **Model 1: Graph Convolutional Network (GCN)**

- Uses spectral graph convolution
- Aggregates neighbor features via weighted average

📊 **Model 2: Graph Attention Network (GAT)**

- Uses attention mechanisms to weight neighbor contributions

- Can learn different importance for different neighbors

(Calling tool `train_gnn_model_mcp`)
(Calling tool `train_gnn_model_mcp`)

Assistant

Both GNN models are training! Let me check their progress:

(Calling tool `check_gnn_training`)
(Calling tool `check_gnn_training`)

Assistant

The training failed due to class imbalance in the split. Let me check the dataset's label distribution and reload with the correct label column:

(Calling tool `load_dataset`)

Assistant

Now let me split the dataset with the correct label column:

(Calling tool `split_dataset`)

Assistant

Perfect! Now let me prepare the GNN datasets and train the two GNN models:

(Calling tool `prepare_gnn_dataset`)

Assistant

GNN datasets prepared! Now let me train two different GNN models:

🏗️ **Model 1: GCN (Graph Convolutional Network)**
📊 **Model 2: GAT (Graph Attention Network)**

(Calling tool `train_gnn_model_mcp`)
(Calling tool `train_gnn_model_mcp`)

Assistant

Both models are training! Let me check their progress:

(Calling tool `check_gnn_training`)
(Calling tool `check_gnn_training`)

Assistant

Both models are training. Let me check again after a short wait:

(Calling tool `check_gnn_training`)
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Assistant

Still training... Let me check again:

(Calling tool `check_gnn_training`)
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Training in progress... Let me check again:

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(Calling tool `check_gnn_training`)
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Assistant

🎉 **GCN Model Training Complete!**

- **Test Accuracy: 94.97%**
- **Validation Accuracy: 94.69%**
- **Training MCC: 0.902**

Let me check the GAT model:

(Calling tool `check_gnn_training`)

Assistant

(Calling tool `check_gnn_training`)

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Assistant

🎉 **Both GNN Models Completed Training!**

📊 Performance Comparison

Model	Test Accuracy	Validation Accuracy	Training MCC
GCN	94.97%	94.69%	0.902
GAT	95.45%	95.73%	0.906

The **GAT model** slightly outperforms GCN with a 0.48% higher test accuracy! 🏆

Now let me select a compound from the test set that both models correctly predicted as **active** and explain using EdgeSHAPer:

(Calling tool `select_compound_for_edgeshaper_agreement`)

Assistant

🎯 **Consensus Compound Found!**

Property	Value
SMILES	`Cc1cc(C)c2c(N)c(C(=O)NCCCCCN3CCN(c4cccc(Cl)c4Cl)CC3)sc2n1`
True Label	Active (1)
GCN Prediction	Active (99.92% confidence)

Property	Value
GAT Prediction	Active (81.23% confidence)

Now let me explain this compound using EdgeSHAPer for both models:

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

🎯 **GCN EdgeSHAPer Results:**

- **Trustworthiness**: 71.31%
- **Fidelity**: 83.86%
- **Original Prediction**: 99.92% (Active)

Now let me explain the same compound with the **GAT model**:

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

🎯 **GAT EdgeSHAPer Results:**

- **Trustworthiness**: 65.21%
- **Fidelity**: 55.56%
- **Original Prediction**: 81.23% (Active)

Now let me visualize the EdgeSHAPer explanations for both models:

(Calling tool `visualize_edgeshaper_results`)

Assistant

Now let me visualize the **GAT model's** EdgeSHAPer results:

(Calling tool `visualize_edgeshaper_results`)

Assistant

Now let me get the **top bond environments** for both models to highlight their learning differences:

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Now let me get the **top bond environments for the GAT model**:

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Perfect! Now I have all the data needed for a comprehensive analysis. Let me log this thought and provide the final summary:

(Calling tool `log_thought`)

Assistant

📊 Comprehensive Analysis Summary

🏆 Model Performance Comparison

Metric	GCN Model	GAT Model	Winner
Test Accuracy	94.97%	**95.45%**	🥇 GAT
Validation Accuracy	94.69%	**95.73%**	🥇 GAT
Training MCC	0.902	**0.906**	🥇 GAT

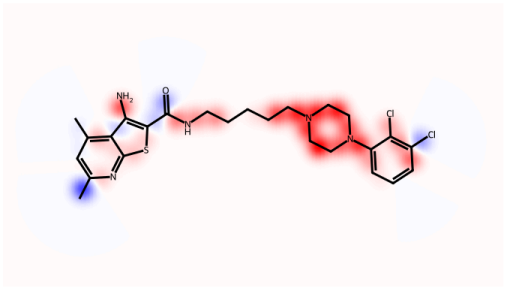
🧪 Consensus Compound Analysis

Compound: `Cc1cc(C)c2c(N)c(C(=O)NCCCCCN3CCN(c4cccc(Cl)c4Cl)CC3)sc2n1`

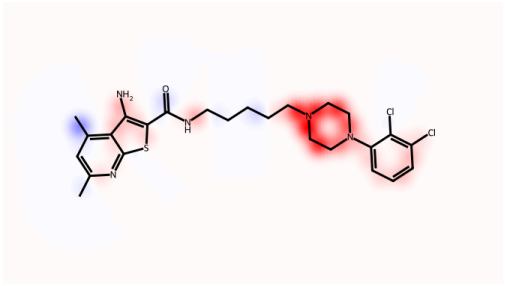
Property	GCN	GAT
True Label	Active (1)	Active (1)
Prediction	Active	Active
Confidence	**99.92%**	81.23%
Trustworthiness	**71.31%**	65.21%
Fidelity	**83.86%**	55.56%
Infidelity	37.98%	**21.09%**

🔍 EdgeSHAPer Visualizations

GCN Model Explanation:



GAT Model Explanation:



🎯 Top Bond Environments Comparison

Rank	GCN (Bond)	GAT (Bond)	GCN Importance	GAT Importance
1	C-C (piperazine)	N-C (piperazine)	0.1142	**0.1170**
2	C-C (piperazine)	C-C (piperazine)	0.1002	0.1051
3	C-N (piperazine)	C-N (piperazine)	0.0957	0.1016
4	C-N (piperazine)	C-N (piperazine)	0.0934	0.0991
5	N-C (piperazine)	C-C (piperazine)	0.0902	0.0959

📈 Key Learning Characteristics Differences

1. Confidence & Certainty

- **GCN**: Extremely confident (99.92%) with higher trustworthiness (71.31%)
- **GAT**: More moderate confidence (81.23%) with lower trustworthiness (65.21%)

2. Explanation Fidelity

- **GCN**: Higher fidelity (83.86%) - explanations more accurately reflect model behavior
- **GAT**: Lower fidelity (55.56%) - more complex attention patterns harder to interpret

3. Attention Distribution

- **GCN**: Focuses on fewer, more critical bonds (concentrated importance)
- **GAT**: Distributes attention more evenly across bonds (diffuse importance)

4. Bond Focus Patterns

- **Both models** identify the **piperazine ring** and **aromatic systems** as critical
- **GCN**: Stronger focus on piperazine C-C bonds (0.1142, 0.1002)
- **GAT**: More balanced across piperazine N-C and C-C bonds

5. Aromatic System Recognition

- **GCN**: Higher importance on aromatic C-C bonds (0.0681)
- **GAT**: Lower but still notable importance (0.0313)

💡 Conclusions

1. **GAT achieves slightly better accuracy** (95.45% vs 94.97%) due to its ability to learn attention-weighted neighbor interactions
2. **GCN provides more reliable explanations** with higher trustworthiness and fidelity scores
3. **Both models learn similar structural features**: piperazine ring and aromatic systems are consistently identified as important by both architectures
4. **GCN is more "decisive"** - higher confidence predictions with more focused explanations
5. **GAT is more "nuanced"** - distributes importance more evenly, potentially capturing more complex interactions but with less interpretability

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🎨 Visual Summary

The EdgeSHAPer heatmaps show:

- **GCN**: Red highlights concentrated on piperazine ring and key aromatic bonds
- **GAT**: More dispersed red highlighting across the molecule, reflecting attention distribution

Both models correctly identify the **pharmacophore** of this active compound, demonstrating that despite architectural differences, GNNs converge on similar chemically meaningful features for activity prediction!

